

# A Study on the Efficient Tracking of ERD based on the Adaptive Identification of the Subject's Reactive Band

M. M. C. Ștefan, I. E. Nicolae, R. Strungaru, G. M. Ungureanu  
Politehnica University of Bucharest  
Bucharest, Romania  
miruna.mc.stefan@gmail.com

M. Vasile, O.A. Băjenaru  
Carol Davila University of Medicine and Pharmacy,  
Neurology Clinic Department, Emergency University  
Hospital Bucharest  
Bucharest, Romania

**Abstract**— Imagining a certain movement produces changes in the subject's EEG motor rhythm in the same manner in which the real execution of the movement does. These variations are reflected by Even-Related Desynchronizations (ERD) and appear as a magnitude decrease of the frequency components comprising the  $\mu$  band. By monitoring the amplitude and power changes, the automatic detection of imagery or movement can be conducted. Furthermore, features can be extracted and used for real-time classification of these events. It is known that in most cases, the majority of changes are constrained within a narrow frequency interval representing the subject's reactive band. This provides means to increase the efficiency of interpreting desynchronizations by working only with the most representative information. The paper presents three improved methods of adaptively modelling the EEG signal, based on a Band Limited Multiple Fourier Linear Combiner (BMFLC). The time-frequency decomposition thus obtained is subsequently exploited in order to automatically identify the subject's specific reactive band. This process is conducted by detecting the highest decrease in power with the purpose of using a selection of power information as features for the classifier. The final goal of the investigated algorithms is to allow discrimination between left and right hand motor imagery, as demanded by a brain computer interface rehabilitation system.

**Keywords**—motor imagery;  $\mu$ -rhythm; ERD detection; adaptive estimation; BCI; BMFLC; time-frequency map; reactive band; optimization

## I. INTRODUCTION

The study of EEG waves has received increasing attention over the past 20 years as it led to a better understanding of the of brain functioning and supported the development of innovative technologies and applications. An example of such advances are Brain Computer Interfaces (BCI) which have found use in various domains with remarkable progress in the area of rehabilitation. One of the most common use of BCI systems is aiding the recovery of motor functions after the responsible cortical areas have been affected by an accident or a stroke [1].

Movement and motor imagery are events that induce changes in the amplitude of EEG signals. More specifically, a simultaneous suppression of neural activity takes place in the

motor area responsible for the task which translates into a decrease in signal amplitude at certain frequencies. These fluctuations are called Event Related Desynchronizations (ERD) and are evaluated in order to distinguish between right and left hand motor imagery [2]. However, these changes in amplitude, more effectively evaluated as changes in power, are not spread out over the entire motor rhythm but occur within a limited sub-band. Thus, rather than using the entire  $\mu$  band, spanning between 6 and 14Hz, it is more efficient to consider as classification features the power values determined over the subject specific reactive 2Hz sub-band. [3] [4]

Commonly used methods for frequency analysis include STFT (Short Time Fourier Transform), which has a low frequency resolution, and the wavelet transform which solves this issue at the cost of high computational requirements, which is not adequate for BCIs. The proposed method, BMFLC (Band-limited Multiple Fourier Linear Combiner), uses Fourier series to model a given signal. It constructs a precise time-frequency decomposition that provides easy access to the signal's power for any frequency at any point in time [3] [5] [6]. The algorithm before improvement has unreliable performance, because it includes a basic Least Mean Square (LMS) routine known to be very sensitive to the chosen parameter values [7]. However, by taking advantage of its great flexibility, also stemming from the LMS, modifications and additions can be made in order to greatly improve the obtained performance, making it compatible with a BCI system [4].

This study evaluates proposed variations of the BMFLC considering the quality of the rendered model, its subsequent utility in ERD detection and accuracy in reactive band identification. The improvement brought by working solely with the reactive band is also discussed with special emphasis placed on the real-time algorithm that can be successfully integrated into a BCI system.

## II. DATASET DESCRIPTION

Part of the BCI competition series, in 2008 the research team at Graz University of Technology recorded data from 8 subjects performing 4 types of motor tasks. The 22 EEG electrodes were placed according to the 10-20 international system and the measurements were conducted monopolarly.

After seating the subjects comfortably in a chair, clues for the four classes of movement were provided. For this study only those trials related to upper limb movement were selected and processed (right hand, left hand) [8].

All algorithms were implemented in MATLAB, including importing and pre-processing the data by the means of the open source EEGLAB toolbox with BioSig extension. A high order FIR filter (500) was used to select the mu band without affecting the phase of the signal. The order of the filter must be chosen considering the sampling frequency and acquisition conditions. In this case, the high order is acceptable because the filter loading period overlaps an EOG evaluation period present at the beginning of each recording. It is important that the EEG is filtered before segmentation to avoid data loss. Right and left hand trials were extracted from the continuous data. To align the trials, the presentation of the cue was chosen as the time-locking event, making each extracted trial 6 seconds long: 2 seconds preceding the cue followed by a 4 seconds period.[8]

### III. METHODS

#### A. Band Limited Multiple Fourier Linear Combiner

The EEG signal is a very complex, nonstationary signal [9] and the BMFLC method assumes that at any point in time its value can be expressed as the sum of instantaneous values rendered by simple sinusoid functions of various frequencies. It attempts to determine the amount in which each sine and cosine component contributes at each moment in time [3], [6]. The LMS is an adaptive steepest descent gradient algorithm used to iteratively minimize the square error. Its wide range of applications is justified by how simple and computationally efficient it has proven to be. The main disadvantages are its sensibility to initialization values and the problems in reaching convergence when noise affects the data [5] [4].

The band of interest  $[\omega_1-\omega_n]$  is divided applying the frequency step  $\Delta \omega$  into  $n$  sub-bands, to form the BMFLC. The  $p^{\text{th}}$  frequency  $\omega_p$  is calculated as  $\omega_p=2\pi f/f_s$  where  $f_s$  is the sampling frequency and  $f$  the corresponding frequency (Hz). In equation (1)  $s_k$  denotes the value of an incoming EEG signal at sampling time  $k$  which ranges within the interval  $[1, m]$ ; the model obtained using this method is  $y_k$ . For frequency  $\omega_p$ , at sampling instant  $k$ ,  $a_{pk}$  and  $b_{pk}$  represent the adaptive weights of the model.

$$y_k = \sum_{p=1}^n a_{pk} \sin(\omega_p k) + b_{pk} \cos(\omega_p k) \quad (1)$$

Let the column vectors  $x_k$  be the input reference vector and  $w_k$  be the adaptive weight vector, at time instance  $k$ .

$$x_k = [\sin(\omega_1 k) \dots \sin(\omega_n k) \cos(\omega_1 k) \dots \cos(\omega_n k)]^T$$

$$w_k = [a_{1k} \ a_{2k} \dots a_{nk}, \ b_{1k} \ b_{2k} \dots b_{nk}]^T \quad (2)$$

The steps followed by the algorithm are:

$$y_k = w_k^T x_k \quad (3)$$

$$\varepsilon_k = s_k - y_k \quad (4)$$

$$w_{k+1} = w_k + 2\eta \varepsilon_k x_k \quad (5)$$

The absolute weight vector  $w_k^f$  encompasses all frequency components at sampling instant  $k$  and is computed accordingly to equation (6).

$$w_k^f = \left[ \frac{a_{1k}^2 + b_{1k}^2}{2} \dots \frac{a_{nk}^2 + b_{nk}^2}{2} \right] \quad (6)$$

Pertaining to the entire signal, the weight matrix  $D$  which represents the time frequency decomposition is expressed as:

$$D = [w_1^f \dots w_{k...}^f \ w_m^f] \quad (7)$$

This matrix offers easy access to the exact values regarding all frequency components. Here,  $D(k,f)$  is the amplitude of frequency component  $f$  for sample  $k$ . The signal's power can immediately be calculated as presented in (8). Thus, by squaring each element of the matrix  $D$  the power distribution  $P$  can be computed for all frequency components. [4] [3].

$$P(k, f) = D(k, f) \times D(k, f) \quad (8)$$

#### B. Normalized least mean square

Normalizing the presented LMS routine (Normalized Mean Square – NLMS) has the purpose of reducing its sensitivity to the scaling of the input which causes convergence problems [10]. In this version, equation (8) replaces (5), and the adjustment amount is normalized to the input power.

$$w_{k+1} = w_k + \frac{2\eta \varepsilon_k^T x_k}{\|x_k\|^2} \quad (9)$$

#### C. Small Laplacian Filter

The signals recoded by each electrode are a sum of the potentials generated by the area directly underneath the electrode as well as the potentials produced by the neighboring areas. Thus, it is advisable to eliminate the electrical activity generated by irrelevant cortical areas along with the continuous component. This is accomplished by subtracting the mean value of the potentials collected by corresponding neighboring electrodes from the EEG channels of interest. The channels of interest are located over the left (C4) and right (C3) regions of the motor cortex (contralateral). For each of the two, the set of

four nearest neighbor electrodes is comprised of those placed closest on a vertical (above, below,) and horizontal axis (on the left and right) [11]. As defined by McFarland et al, this small Laplacian filter is described by equation (9).

$$V_{CH}^{LAP} = V_{CH}^{ref} - \frac{1}{4} \sum_{j \in SCH} V_j^{ref} \quad (10)$$

where CH is the location of interest and SCH the set of four nearest neighbor electrodes.

#### D. Online: Forcefully Adjusting the Weights

This method adds another step to the algorithm in which the absolute value of the previous iteration's error  $\epsilon_{k-1}$  is compared to a threshold value ( $\epsilon^h$ ). If the error level is exceeded, the weights are decreased by the inverse of a calibration factor  $c_f$ , according to equation (11) [4].

$$w_k = w_k - \frac{1}{c_f} \times w_k \quad (11)$$

#### E. Offline: Adding a learning step

The moment of desynchronization produces large errors which can be overcome if the algorithm is given more time to adapt. The same effect can be accomplished by repetitively applying the method on windows of a certain length rather than the entire trial. The number of repetitions is inversely proportional to the length of the selected window. The matrix D is obtained by concatenating the weights rendered by each of the training periods [4].

#### F. Determining the Subject's Reactive Band

Because the variation is the aspect of interest, working strictly with the frequencies comprising the active band which contains the highest difference in signal power can be viewed as a manner of increasing the Signal-to-Noise ratio. The average power for the imagery reference period [3.5s, 5s] is subtracted from the rest reference period [0.5s, 1.5s], yielding a value for each frequency in the mu band [3]. These values are summed together in frequency bands of constant width. The sub-band that gives the most important difference in power is considered the subject's reactive band.

$$P_{rest} = \frac{\sum_{k=k1}^{k2} P(k, f)}{k2 - k1} \quad (12)$$

$$P_{imag} = \frac{\sum_{k=k3}^{k4} P(k, f)}{k2 - k1} \quad (13)$$

$$P_{diff} = P_{rest} - P_{imag} \quad (14)$$

However, the qualitative evaluation of the chosen sub-band must be conducted. This selection is significant only if, by comparison with the excluded frequencies, the chosen sub-band brings a relevant improvement. In order to investigate this aspect, the parameter  $\Delta$  is calculated as follows [3].

$$\Delta = \frac{A_{opt}}{A_{\mu}} \cdot 100 \quad (15)$$

It evaluates the positive area underneath the power difference curve in the said interval ( $A_{opt}$ ) and matches it to the positive area existing in the entire mu band ( $A_{\mu}$ ). The higher the value obtained, the more relevant the selected band is.

#### G. ERD Detection

Detection of the ERD is essential for the functioning of a BCI system. The value of the desynchronization is calculated for every sample by summing up the values of the signal's power over the considered frequencies. These sums are compared with the time average of the power sum value calculated over a certain interval considered of reference. In this situation, the time-frame is the rest reference period mentioned earlier.

$$P_R(k) = \sum_{f=f1}^{f2} P(k, f) \quad (16)$$

$$P_{ref} = \frac{\sum_{k=k1}^{k2} P_R(k, f)}{k2 - k1} \quad (17)$$

$$ERD(k) = \frac{P_R(k) - P_{ref}}{P_{ref}} \quad (18)$$

The reason why we seek to work solely with the subject's reactive band is to increase the percentage of the power difference with respect to the power reference which is influenced by the chosen frequency interval. [3]

## IV. RESULTS

Each trial is separately modeled using the BMFLC and the final matrix **D** for a certain subject is obtained by averaging the resulting weight matrices. The %accuracy of the model is used to evaluate each method:

$$\%Accuracy = \frac{RMS(s) - RMS(\epsilon)}{RMS(s)} \quad (19)$$

The root mean square is calculated as  $RMS(s) = \sqrt{\sum_{i=1}^n s_i^2 / n}$ , n being the number of trials per subject.

The Laplacian filter is applied before any other methods to enhance Signal-to-Noise ratio; the variations are emphasized

and the improvement is obvious when observing the time-frequency decomposition matrix portrayed as an image [11].

The offline methods (section E) have been implemented using half second windows with 10 repetitions and one second long windows with 5 repetitions respectively. The time needed for the algorithm to converge on simulated, noisy, data composed of 6 sine functions is approximately 5 seconds. Thus the window length has been matched to the number of repetitions as so to yield a total runtime of 5 seconds. Error values are highest when passing from one windows of repetition to the next but decrease rapidly as the algorithm adapts. In terms of modelling accuracy using half-second has better performance, but when detecting the active band is considered, the one second window proves more reliable. Active band results obtained in [3] were used as reference for the evaluation of the presented algorithms. In both cases this method produces fake large values at the ends of the mu band which are insignificant when working with high quality data but prove more damaging with the increase of noise.

TABLE I. OFFLINE METHOD: PERFORMANCE COMPARISON OF THE TWO WINDOWS AND THE NORMALISED VERSION

| Method             | % Accuracy |          |        |          |
|--------------------|------------|----------|--------|----------|
|                    | Normalized |          | Simple |          |
|                    | Mean       | $\sigma$ | Mean   | $\sigma$ |
| Half-Second Window | 83.08      | 2.58     | 94.35  | 1.11     |
| One Second Window  | 75.50      | 4.09     | 89.98  | 1.45     |

The online method is especially important as its performance proves only a few percents weaker than those obtained using the proposed offline processing algorithms. The conclusion is drawn observing the mean percentual accuracy and standard deviation ( $\sigma$ ) presented in Tables I and IV. For each value of threshold (th) there is an ideal value for cf (as presented in Section D, (11)). Both smaller and larger values of cf lead to a decrease in model accuracy. The obtained performances are not dependent on the subject's particularities.

Two different behaviors can be delimited with respect to performance for both high and low quality data. If the error threshold has larger values (10-2 – 10-4), cf must take larger values decreasing the subtracted amount. In this case, the model's accuracy percent will be very high for good data but very small, even negative, for noisy cases. On the other hand, if th has very small values (10-4 -10-7), cf must as well assume smaller values. It can be chosen so as to favor noise corrupted recordings, making the method very robust to noise at the cost of a 20 to 30 % decrease in accuracy.

This latter version has been proven to flatten variations with respect to frequency. Fluctuations appearing in the Pdiff curve are diminished preventing the accurate active band identification based on maximum power difference and canceling the improvement normally brought in ERD evaluation.

Normalization keeps all accuracy percent values above zero by not allowing the error value to surpass the input signal. For the offline algorithms, it reduces performance by amplifying

errors. In the case of the online versions, percent accuracy values are very low because, while the model manages to retain signal changes, it covers a smaller amplitude range. Important variations are kept and emphasized by using NLMS and the method behaves better in identifying the reactive band than the simple variant.

TABLE II. ONLINE METHOD: OPTIMAL PARAMETER COMBINATIONS

| th        | cf                          |                                            |
|-----------|-----------------------------|--------------------------------------------|
|           | Favoring global performance | Favoring performance for high quality data |
| 0.02      | -                           | 150                                        |
| 0.01      | -                           | 130                                        |
| 0.007     | -                           | 130                                        |
| 0.001     | -                           | 100                                        |
| 0.0005    | -                           | 100                                        |
| 0.0001    | 5                           | 70                                         |
| 0.00001   | 5                           | 10                                         |
| 0.000001  | 5                           | 15                                         |
| 0.0000001 | 5                           | 15                                         |

Global performance refers to overall performance for high quality and noisy data. For fields marked “-“global performance cannot be considered as the percentual accuracy is very low.

TABLE III. ONLINE METHOD: % ACCURACY FOR PARAMETER COMBINATIONS PRESENTED IN TABLE I

| th        | % Accuracy |          |       |          |
|-----------|------------|----------|-------|----------|
|           | Mean       | $\sigma$ | Mean  | $\sigma$ |
| 0.02      | -          | -        | 82.60 | 4.72     |
| 0.01      | -          | -        | 84.94 | 3.24     |
| 0.007     | -          | -        | 85.79 | 3.21     |
| 0.001     | -          | -        | 88.06 | 2.40     |
| 0.0005    | -          | -        | 88.24 | 1.26     |
| 0.0001    | 73.08      | 1.04     | 88.89 | 1.23     |
| 0.00001   | 73.07      | 1.07     | 81.40 | 1.54     |
| 0.000001  | 73.08      | 1.07     | 85.55 | 1.49     |
| 0.0000001 | 73.09      | 1.07     | 85.55 | 1.49     |

TABLE IV. OVERVIEW: MEAN PERFORMANCE OF ALL METHODS

| Method             | % Accuracy |          |
|--------------------|------------|----------|
|                    | Mean       | $\sigma$ |
| On-line            | 88.89      | 1.23     |
| One Second Window  | 94.35      | 1.11     |
| Half-Second Window | 89.98      | 1.45     |

In all cases that investigate reliable data, an improvement in ERD detection is obtained by applying all of the proposed methods. The enhancement can be observed on the ERD curve

as an increase in level difference between the two states when using solely the subject's active band.

Considering that a BCI system must respond promptly and the limitations of computational resources, simplicity is a key attribute in choosing the classification algorithms, thus the selected options are LDA and QDA [12]. A different classifier was designed and trained for each subject because the functioning of every person is unique and so are the results obtained.

Considering that a BCI system must respond promptly and The features used are the power for each frequency in the considered band, averaged over time for both the left and right hand actions [3]. The most appropriate decision window spans over the [3s, 4s] interval and the period considered here was 0.5 seconds long starting at t=3s. The feature vector follows the structure presented in (21). In equation (20),  $P(k, f)$  is the power of frequency component  $f$  at sampling instant  $k$  first mentioned in (8), and  $N$  is the number of samples corresponding to the time interval considered, in this case 125.

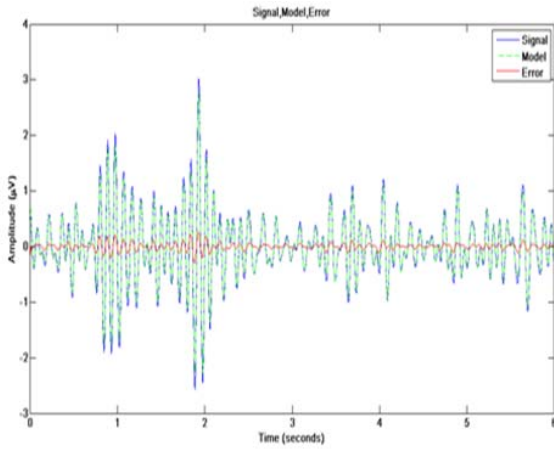


Fig. 1. The model obtained using the on-line method on a single left hand imagery trial for subject 2.

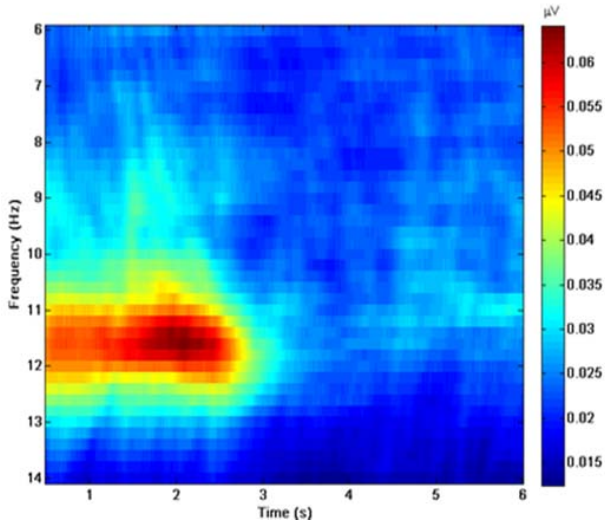


Fig. 2. Weight matrix as colormap averaged for subject 2, left hand motor imagery. The method used is D. which treats the entire trial.

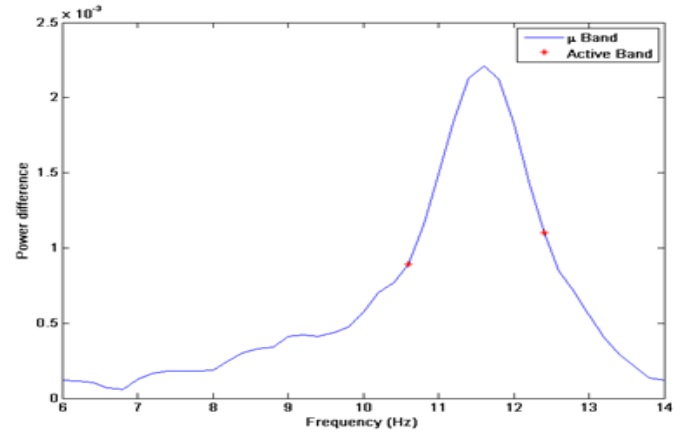


Fig. 3. Power difference PDiff between the two states and the identified active band marked.

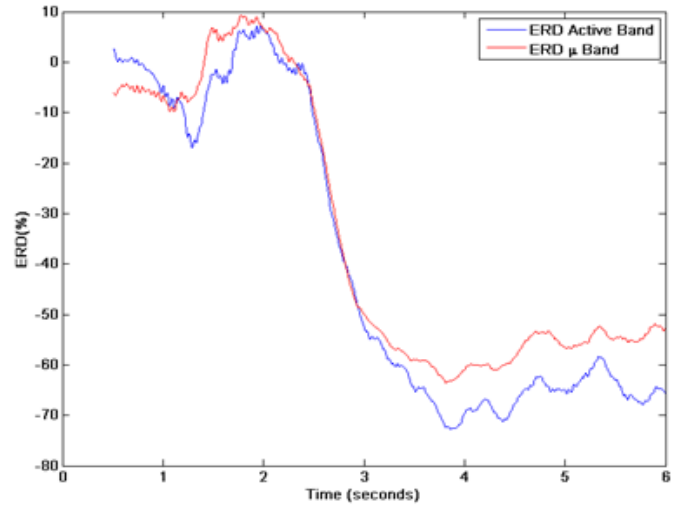


Fig. 4. ERD computed in both the entire mu band and the active band

$$p_i^f = \frac{1}{N} \sum_{k=1}^{i+N} P(k, f) \quad (20)$$

$$F_{BMFLC}^i = (p_{i,C3}^{f_{\min}} \dots p_{i,C3}^{f_{\max}}, p_{i,C4}^{f_{\min}} \dots p_{i,C4}^{f_{\max}}) \quad (21)$$

The discrimination results for the online algorithm discussed here are presented in Table VI, the error mentioned being the total misclassification error. Using the active band proves less efficient than considering the entire mu band, when applying the LDA classification to the selected data. Most likely the separation surface delimiting right and left hand imagery obtained in this manner is not linear. In contrast, the QDA conforms to the expected behavior and has a significantly lower error when working solely with the active band. If we are to consider the balance in between the performance obtained and the number of featured used, the linear classifier working with mu band features requires 4 times more information than the quadratic classifier applied upon active band features.

TABLE V. REACTIVE BAND IDENTIFIED WITH THE ONLINE METHOD

| Subject | Hand  | On-line - Simple |      |       | On-line - Normalized |      |          | Offline: Half-second window |      |          | Offline One second window |      |          | Reference values |      |          |
|---------|-------|------------------|------|-------|----------------------|------|----------|-----------------------------|------|----------|---------------------------|------|----------|------------------|------|----------|
|         |       | Inf              | Sub  | Hand  | Inf                  | Sup  | $\Delta$ | Inf                         | Sup  | $\Delta$ | Inf                       | Sup  | $\Delta$ | Inf              | Sup  | $\Delta$ |
| 1       | Left  | 7                | 8.8  | 29.84 | 7                    | 8.8  | 29.84    | 12.2                        | 14   | 36.01    | 7                         | 8.8  | 28.46    | 7                | 9    | 32.20    |
| 1       | Right | 10.8             | 12.6 | 46.27 | 10.8                 | 12.6 | 46.27    | 11                          | 12.8 | 50.82    | 10                        | 12.2 | 41.65    | 10               | 12   | 43.20    |
| 2       | Left  | 10.6             | 12.4 | 61.42 | 10.6                 | 12.4 | 61.42    | 10.8                        | 12.6 | 64.86    | 10.4                      | 12.2 | 56.47    | 11               | 13   | 32.50    |
| 2       | Right | 10.8             | 12.6 | 68.06 | 10.8                 | 12.6 | 68.06    | 10.8                        | 12.6 | 73.69    | 10.4                      | 12.2 | 64.00    | 11               | 13   | 26.80    |
| 3       | Left  | 6                | 7.8  | 41.64 | 6                    | 7.8  | 41.64    | 6                           | 7.8  | 60.72    | 6                         | 7.8  | 61.65    | 10               | 12   | 36.90    |
| 3       | Right | 6                | 7.8  | 40.10 | 6                    | 7.8  | 40.10    | 6                           | 7.8  | 52.6     | 6                         | 7.8  | 51.23    | 10.4             | 12.4 | 76.60    |
| 4       | Left  | 6                | 7.8  | 43.44 | 6                    | 7.8  | 43.44    | 6                           | 7.8  | 56.34    | 6                         | 7.8  | 56.17    | 0                | 0    |          |
| 4       | Right | 6                | 7.8  | 33.14 | 6                    | 7.8  | 33.14    | 6                           | 7.8  | 45.314   | 6                         | 7.8  | 45.53    | 9                | 11   | 63.60    |
| 5       | Left  | 9.6              | 11.4 | 72.44 | 9.6                  | 11.4 | 72.44    | 8.6                         | 10.4 | 39.8     | 9.8                       | 11.6 | 54.85    | 11               | 13   | 78.80    |
| 5       | Right | 8.4              | 10.2 | 37.92 | 8.4                  | 10.2 | 37.92    | 12                          | 13.8 | 59.08    | 12.2                      | 14   | 34.18    | 11               | 13   | 59.70    |
| 6       | Left  | 9.2              | 11   | 46.03 | 9.2                  | 11   | 46.03    | 12.2                        | 14   | 29.93    | 9.6                       | 11.4 | 36.78    | 9                | 11   | 51.60    |
| 6       | Right | 9.6              | 11.4 | 47.55 | 9.6                  | 11.4 | 47.55    | 10.8                        | 12.6 | 34.34    | 9.6                       | 11.4 | 38.95    | 9                | 11   | 54.40    |
| 7       | Left  | 9.2              | 11   | 45.21 | 9.2                  | 11   | 45.21    | 6                           | 7.8  | 48.93    | 6                         | 7.8  | 43.42    | 9                | 11   | 85.10    |
| 7       | Right | 8.4              | 10.2 | 42.38 | 8.4                  | 10.2 | 42.38    | 6                           | 7.8  | 48.93    | 6                         | 7.8  | 48.59    | 9                | 11   | 74.10    |
| 8       | Left  | 9.8              | 11.6 | 73.48 | 9.8                  | 11.6 | 73.48    | 9.8                         | 11.6 | 47.6     | 9.6                       | 11.4 | 56.72    | 9.8              | 11.8 | 83.60    |
| 8       | Right | 10.2             | 12   | 58.10 | 10.2                 | 12   | 58.10    | 10.8                        | 12.6 | 45.43    | 11.2                      | 12   | 53.39    | 10.2             | 12.2 | 70.50    |

"Inf" & "Sup"- Inferior and superior limits of the identified reactive band

In this particular case, with a 0.2 Hz frequency resolution, a set of 82 values is compared to a set of 20 values, with a mean performance improvement of only 5%.

TABLE VI. ERROR RESULTS OBTAINED BY USING THE LDA AND QDA ON BOTH TYPES OF DATA

| Subject | LDA    |        | QDA    |        |
|---------|--------|--------|--------|--------|
|         | Active | $\mu$  | Active | $\mu$  |
| 1       | 0.2319 | 0.0652 | 0.0870 | 0.5000 |
| 2       | 0.0949 | 0.0073 | 0.0584 | 0.5037 |
| 3       | 0.3597 | 0.0863 | 0.1439 | 0.5180 |
| 4       | 0.3023 | 0.0465 | 0.1705 | 0.4884 |
| 5       | 0.2566 | 0.0177 | 0.0885 | 0.4956 |
| 6       | 0.3083 | 0.0602 | 0.0677 | 0.5038 |
| 7       | 0.2077 | 0.0462 | 0.0923 | 0.4923 |
| 8       | 0.0948 | 0.0000 | 0.0259 | 0.4569 |

## V. CONCLUSIONS

After applying the basic, unimproved BMFLC (A.), presented in this paper and also discussed in [3], on a simple simulated signal consisting of six added sinusoids, we noticed that the modeling algorithm fails to reach convergence in reasonable time, needing up to 5 seconds to stabilize. In order to make this method appropriate for reactive band identification and for use in BCI, some changes have been considered, which resulted into one online (D.) and one

offline(E.) versions of the algorithm. These methods successfully overcome the high errors and important stationarity losses occurring at the moment of desynchronization in a complex EEG signal. Their behavior has been studied in both simple and normalized (B.) cases and this paper provides an evaluation of their performance judging both the quality of the model and the success in correctly identifying the subject's reactive band. The data used in our experiment is tested and readily available as part of the fourth BCI competition, and the results obtained are comparable and concordant with those presented in other works addressing the matter.

The presented results recommend the use of the online method for real time BCI. Best active band identification is obtained by applying the normalized version of the algorithm. This is explained by the apparent amplification of significant variations that happens when the amplitude range is narrowed and the changes (in time) of the signal are respectively scaled. On the other hand, if signal amplitude is relevant, for example used as a trigger for delivering feedback, then the simple variant is best used. Both situations are computationally efficient, easy to implement and deliver the result in a timely manner. For example, up to 4 classification procedures can be conducted within a single second with a sampling frequency of 250Hz and the number can increase with a higher sampling frequency. A promising alternative to the presented methods which could potentially improve the performance obtained is the adaptive modelling algorithm using a Kalman filter.

In terms of ERD detection and active band identification, we believe more parameters are needed to evaluate the

obtained results. The placement and shape of the identified active band, observable when plotting the PDiff curve, and the presence of oscillations over the excluded interval are important factors that indicate a result is incorrect regardless of power ratio values. These cues appear in cases of noisy data, for example if we examine the models built with the offline algorithm for subjects 3,4 and 7. From the results described in Table V. it can be noticed that the active band has been placed at the lower limit of the  $\mu$  interval, and when plotting PDiff we observed that it presents the respective oscillations.

The reactive band identification can be designed as a quick calibration procedure followed before all other tasks that would allow a more efficient interaction with the BCI system for rehabilitation purposes.

The field of BCI is an expanding domain which finds more and more applications in various directions (for example command and control). Given the results presented in this paper and the existing international development we strongly believe the matter is worth exploring in more detail with the purpose of obtaining better performance and deepening our understanding of the human brain.

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